ASSIGNMENT 01 – PFP191

Duration: 70'

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Software Requirements

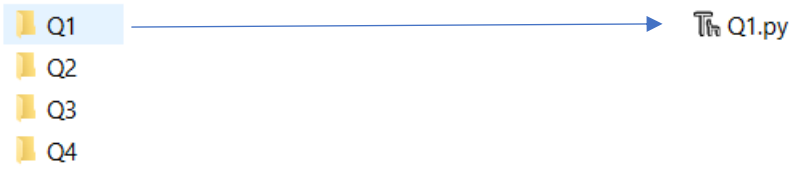
- Thonny, Notepad, Command Prompt, WinRAR / WinZip (or compress to ZIP file of Windows), Windows Explorer (File Explorer) on Windows 7 and above.

Students are ONLY Allowed to use:

- The materials like sample codes and program examples are stored on computer only.

Instructions

- Step 1: Students download the given materials from PEA Client.
- Step 2: Students read questions and prepare answers in the given template.
- Step 3: Submit a solution for each question:
    - The result is one folder for each question (Q1,Q2,Q3,Q4…), which contains **.py** source files.
      Example: Folder "Q1" contains Q1.py for question 1



    - For each question, you must submit one folder as above to PEA Client.

❖ **Importance:**

- Solutions will be marked by Automated Marking Software.
- **The use of tools other than those allowed in the above section whether intentionally or unintentionally, is considered a violation of the exam rules, and the mark is 0**
- Do not change the names of the folders, files, and, files and structure (format) of .py files specified in the exam. If you change it, the marking software can not find the execute file (.py) or the program output to mark, thus the score will be 0
- Do not edit given statements in the **main** function. If you change, the marking software can not mark and the score will be 0.

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Question 1: (3 marks)

The given file Q1.cpp contains statements to input data for 2 integers a and b, which are 2 sides of a right triangle. You should write statements to print the area and perimeter of the triangle.

Hint:

- The area of a right triangle is (a*b)/2
- The hypotenuse c of a right triangle is calculated as sqrt(a*a + b*b)
- The perimeter of the triangle is a+b+c

Notes:

- The output result is formatted in two decimal places

Sample input and output:

Input: a = 3, b = 4

After processing: 6.00 and 12.00

Output for marking:

OUTPUT:

6.00

12.00

Question 2:

(2 marks)

The given file Q2.cpp contains statements to input data for 3 integers a, b and c. You should write statements to get the smallest and largest number among the 3 numbers.

Sample input and output:

Input: a = 6, b = 5, c = 3

After processing: result : 3 6

Output for marking:

OUTPUT:

3

6

Question 3:

(2 marks)

The given file Q3.cpp contains statements to input data for two integers a, b (b>0). You should write statements to print the result of dividing a by and the remainder of that division.

Sample input and output:

Input: a = 5, b = 2

After processing: 2 1

Output for marking:

OUTPUT:

2

1

Question 4:

(3 marks)

The given file Q4.cpp contains statements to input data for 2 integers r and h, which are the radius and height of the cylinder. You should write statements to print the volume of the cylinder.

Hint:

- You must use PI in math library to get the Pi number.

Notes:

- The output result is formatted in two decimal places

Sample input and output:

Input: a=1, h =5

After processing: 15.71

Output for marking:

OUTPUT:

15.71